



SUPERIOR UNIVERSITY

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Lab 11 – Tasks

Difference Between Key Generate Ai concepts:-

1. **LangChain** A Python framework/library used to build applications powered by LLMs. It helps chain together prompts, memory, tools, retrieval (RAG), agents, etc. Think of it as the “developer toolkit” to create smart AI apps easily.
2. **LLMs (Large Language Models)** Huge neural networks (like GPT-4, LLaMA, PaLM) trained on massive text data to understand and generate human-like text. They are the core “brain” that actually generates answers.
3. **RAG (Retrieval-Augmented Generation)** A technique that combines an LLM with an external knowledge base. Before generating an answer, the system first retrieves relevant documents (using vector search), then feeds them to the LLM so the answer is accurate and up-to-date (solves hallucination problem).
4. **FAISS (Facebook AI Similarity Search)** A high-speed library (developed by Meta) for searching and clustering similar vectors (embeddings). It is commonly used as the search engine inside RAG systems to find the most relevant documents quickly.
5. **Vector** A list of numbers (embedding) that represents the semantic meaning of text, image, or any data in a numerical form. Example: the word “king” → [0.12, -0.45, 0.89, ...]. Similar meanings have similar vectors.

6. **VectorDB (Vector Database)** A specialized database (Pinecone, Chroma, Weaviate, Qdrant, etc.) designed to store millions of vectors and perform fast similarity/nearest-neighbor searches. Used heavily in RAG pipelines.
7. **Generative AI (GenAI)** The broad category of AI that can create new content (text, images, audio, code, video) from scratch by learning patterns from training data. Examples: ChatGPT, DALL-E, Stable Diffusion, Midjourney.
8. **GANs (Generative Adversarial Networks)** A specific type of Generative AI architecture consisting of two networks — Generator (creates fake data) and Discriminator (tries to detect fake vs real). They compete and improve each other. Famous for generating realistic images (e.g., deepfakes, StyleGAN faces).

